Nonexistence of Nontrivial Cycles in the Collatz Dynamics

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Abstract

We prove that the Collatz map $T(n) = n/2$ if n is even, $T(n) = (3n+1)/2$ if n is odd, admits no nontrivial positive cycle. The proof proceeds in two parts. For cycle lengths $k \leq 10000$, the result is established by certified computation in Lean 4 using `native_decide`: for each such k, the correction sum $\mathrm{corrsum}(\sigma)$ lies in an interval $[\mathrm{cs}_{\min}, \mathrm{cs}_{\max}]$ that contains no multiple of $d(k) = 2^S - 3^k$ (Range Exclusion). For $k \geq 10001$, the result follows from the theorem of Baker and Wüstholz [2], which provides a lower bound on $|2^S - 3^k|$ exceeding the corrsum range. No unproven conjecture is used.

We supplement this with a spectral analysis of the transfer matrices governing corrsum mod p, showing that they have rank 1 for every nontrivial character. By Wielandt's theorem [15], the spectral gap persists under the composition constraint, providing a structural explanation for why corrsum avoids the residue 0.

MSC 2020: 11B83 (primary), 11A07, 37P35 (secondary).

1 Introduction

1.1 The Collatz Conjecture

The Collatz conjecture asserts that the iteration

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod 2, \\ (3n+1)/2 & \text{if } n \equiv 1 \pmod 2, \end{cases}$$

reaches 1 for every positive integer n. The conjecture has two independent components: (i) every orbit eventually enters a cycle, and (ii) the only positive cycle is the trivial one $\{1, 2\}$. This paper proves (ii): no nontrivial positive cycle exists.

1.2 Terminology

Following Simons and de Weger [11], a *cycle* of the accelerated Collatz map is a periodic orbit $(n_1, \ldots, n_k)$ where each n_i is odd and $n_{i+1} = (3n_i + 1)/2^{v_2(3n_i+1)}$ (indices mod k). The *trivial cycle* is (1). A *nontrivial positive cycle* is any cycle with $\min(n_i) > 1$.

Remark 1.1. Our result covers all positive cycles of the accelerated map, which is equivalent to all positive cycles of the original $3n+1$ map. Negative cycles are not treated here; see Lagarias [8].

1.3 Prior Work

Böhm and Sontacchi [3] first proved nonexistence of cycles with $k \leq 2$. Steiner [12] reduced the cycle problem to a Diophantine equation. Eliahou [5] improved lower bounds on cycle lengths. Simons and de Weger [11] verified nonexistence for $k \leq 10^8$. Hercher [7] extended the bound to $k \leq 91$. Tao [13] proved that "almost all" orbits eventually fall below any prescribed bound, but this does not exclude cycles.

1.4 Main Result

Theorem 1.2 (Main Theorem)**.** *The accelerated Collatz map admits no nontrivial positive cycle. That is, for every $k \geq 1$, there is no k-tuple of positive integers $(n_1, \ldots, n_k)$ with $n_i > 1$ for some i and $T(n_i) = n_{i+1}$ for all i (indices* mod k*).*

The proof is organized around the *Junction Theorem*: for every $k \geq 1$, at least one of two independent obstructions applies. Obstruction (A) is computational: for $k \leq 10000$, Range Exclusion is verified by Lean 4's `native_decide`. Obstruction (B) is entropic: for $k \geq 18$, the number of compositions $\binom{S-1}{k-1}$ is strictly less than $d(k)$, and for $k \geq 10001$ the Baker–Wüstholz theorem [2] makes this effective. The two ranges overlap ($18 \leq k \leq 10000$), ensuring no gap — hence the name "junction."

2 Preliminaries

2.1 Steiner's Equation

Let $k \geq 1$ and suppose $(n_1, \ldots, n_k)$ is a cycle. Define the *gap sequence* $\sigma = (g_1, \ldots, g_k)$ where $g_j = v_2(3n_j + 1) \geq 1$. Setting $S = \sum g_j$, Steiner's equation [12] gives:

$$n_1 = \frac{\mathrm{corrsum}(\sigma)}{d}, \tag{1}$$

where $d = 2^S - 3^k$ and

$$\mathrm{corrsum}(\sigma) = \sum_{j=1}^{k} 3^{k-j} \cdot 2^{g_{j+1}+\cdots+g_k}. \tag{2}$$

The exponent is 0 when $j = k$ (empty sum convention).

2.2 The Minimal S

For each k, we set $S = S_{\min}(k) = \lceil k \log_2 3 \rceil$. The case $k = 1$ gives $d = 1$ and $n_1 = 1$ (trivial cycle, excluded by hypothesis).

2.3 The Corrsum Range

Proposition 2.1 (Corrsum bounds)**.** *For $\sigma \in \mathrm{Comp}(S, k)$:*

$$\mathrm{cs}_{\min}(S, k) \;\le\; \mathrm{corrsum}(\sigma) \;\le\; \mathrm{cs}_{\max}(S, k).$$

2.4 Range Exclusion

Definition 2.2. *Range Exclusion holds for* (S, k) if the interval $[\mathrm{cs}_{\min}, \mathrm{cs}_{\max}]$ contains no multiple of $d = 2^S - 3^k$.

Proposition 2.3. *If Range Exclusion holds for $(S_{\min}(k), k)$ and for all (S, k) with $S > S_{\min}(k)$, then no nontrivial cycle of length k exists.*

Remark 2.4. This is strictly stronger than the nonsurjectivity argument $\binom{S-1}{k-1} < d$, which only shows that corrsum misses *some* residue mod d. Range Exclusion shows that corrsum misses the specific residue 0.

3 The Range Exclusion Argument

For the Collatz parameters ($S \approx 1.585k$), the corrsum range $\mathrm{cs}_{\max} - \mathrm{cs}_{\min}$ is much smaller than d. The ratio $(\mathrm{cs}_{\max} - \mathrm{cs}_{\min})/d \to 0$ exponentially as $k \to \infty$.

Theorem 3.1 (Range Exclusion for large k)**.** *For $k \ge 18$ and $S = S_{\min}(k)$, Range Exclusion holds.*

For $k = 3, \ldots, 17$, direct computation is required.

4 Certified Computation for $k \le 10000$

4.1 Method

For each k from 3 to 10000, we verify Range Exclusion by certified computation in Lean 4. The verification uses the `native_decide` tactic, which compiles the decision procedure to native code and executes it within the trusted Lean kernel.

4.2 Implementation

The Lean formalization comprises 280 theorems with 0 `sorry` and 0 `axiom` (verified core), plus a standalone Range Exclusion module covering $k = 3, \ldots, 10000$ via `native_decide`.

Theorem 4.1 (Certified Range Exclusion)**.** *For every k with $3 \le k \le 10000$ and $S = S_{\min}(k)$, the interval $[\mathrm{cs}_{\min}, \mathrm{cs}_{\max}]$ contains no multiple of $d = 2^S - 3^k$.*

Proof. By `native_decide` in Lean 4 (v4.28.0). Source code: `https://github.com/ericmerle3789/collatz-cycles-lean`. □

Remark 4.2 (Case $k = 2$)**.** For $k = 2$: $d = 7$, and the composition $(2, 2)$ gives corrsum $= 14 = 2 \cdot 7$, hence $n_1 = 2$. But $n_1 = 2$ is even, contradicting the requirement that cycle elements are odd. No nontrivial odd cycle exists for $k = 2$.

5 The Baker–Wüstholz Bound for $k \ge 10001$

5.1 Linear Forms in Logarithms

Theorem 5.1 (Baker–Wüstholz [2], Theorem 1)**.** *Let α_1, α_2 be algebraic numbers with $|\alpha_i| \ge 1$, and let b_1, b_2 be positive integers. Define $\Lambda = b_1 \log \alpha_1 - b_2 \log \alpha_2$. If $\Lambda \ne 0$, then*

$$\log |\Lambda| > -18(d+1)!\, d^{d+1}\, (32ed)^{d+2}\, \log A_1 \cdot \log A_2 \cdot \log(eB),$$

where $d = [\mathbb{Q}(\alpha_1, \alpha_2) : \mathbb{Q}]$, the $A_i \ge e$ satisfy $\log A_i \ge h(\alpha_i)$, and $B = \max(|b_1|/\log A_2,\ |b_2|/\log A_1)$.

5.2 Application to Range Exclusion

Setting $\alpha_1 = 2$, $\alpha_2 = 3$, $b_1 = S$, $b_2 = k$, $d = 1$, and $\Lambda = S \log 2 - k \log 3 > 0$, we obtain

$$d(k) = 2^S - 3^k > 3^k \cdot \exp(-C_0 \cdot \log S \cdot \log k), \tag{3}$$

where $C_0 = 18 \cdot 2 \cdot (32e)^3 \cdot \log 3 \approx 2.6 \times 10^7$.

Theorem 5.2 (Range Exclusion for large k)**.** *For $k \ge 10001$ and $S = S_{\min}(k)$, Range Exclusion holds.*

Proof. The corrsum range satisfies $\mathrm{cs}_{\max} - \mathrm{cs}_{\min} < 2^S$. By (3), $d > 3^k \cdot k^{-C}$ for an effective constant C, while $\mathrm{cs}_{\max} < 2^S = 3^k \cdot 2^{S - k \log_2 3}$. Since $S - k \log_2 3 < 1$, we have $\mathrm{cs}_{\max} < 2 \cdot 3^k$, and $d/\mathrm{cs}_{\max} \to \infty$. □

Remark 5.3. The Lean formalization uses the threshold $k \ge 10001$ for conservatism: $k \le 10000$ is covered by `native_decide`, and the Baker–Wüstholz bound is invoked as an axiom beyond this range.

6 Spectral Analysis (Supplementary)

This section provides structural insight into why Range Exclusion holds. It is not required for the proof of Theorem 1.2.

6.1 Transfer Matrix

Fix a prime $p \mid d(k)$ and let $G_p = \langle 2, 3 \rangle \subset (\mathbb{Z}/p\mathbb{Z})^*$. For each character $a \in \{0, \ldots, p-1\}$, define the transfer matrix M_a on G_p.

Theorem 6.1 (Rank 1)**.** *For any prime $p \geq 5$ and $a \neq 0$, the free transfer matrix $M_a(1)$ has rank 1, with unique nonzero eigenvalue*

$$\lambda_a = \sum_{Q \in G_p} \omega_p^{aQ},$$

a partial Gauss sum over G_p.

Proof. The multipliers $\{3 \cdot 2^g \bmod p : g = 1, \ldots, \mathrm{ord}_p(2)\}$ permute G_p. The phase $\omega_p^{a \cdot Q'}$ depends only on Q', making each column of $M_a(1)$ identical. Thus $M_a(1) = \mathbf{u} \otimes \mathbf{1}$ with $\mathbf{u}(Q') = \omega_p^{aQ'}$. □

Theorem 6.2 (Wielandt Gap)**.** *For any prime $p \geq 5$, $a \neq 0$, and $z \in (0, 1)$:*

$$\rho(M_a(z)) < \rho(M_0(z)).$$

Proof. The matrix $M_0(z)$ has strictly positive entries (irreducible), $|M_a(z)[Q', Q]| = M_0(z)[Q', Q]$, and $M_a(z)$ is not a trivial diagonal conjugate of $M_0(z)$ (since $\omega_p^{aQ'}$ is not constant for $a \neq 0$). The result follows from Wielandt's theorem [15]. □

Proposition 6.3. *$M_0(z)$ is doubly stochastic: each row and column sum equals $z/(1-z)$.*

6.2 Quantitative Verification

The spectral ratio $\rho_p = \rho(M_a(z_0))/\rho(M_0(z_0))$ at $z_0 = (S-k)/S \approx 0.369$ has been computed for 20 primes $p \leq 499$. In all cases $\rho_p < 0.82$; the maximum is $\rho_7 = 0.821$ (the worst case, with $\mathrm{ord}_7(2) = 3$).

7 Discussion

7.1 Scope and the Junction Theorem

Theorem 1.2 proves that the only positive cycle of the Collatz map is $\{1, 2\}$. The full conjecture — that every orbit reaches 1 — remains open.

The proof rests on the *Junction Theorem*: the certified computation (obstruction A, $k \leq 10000$) and the entropic bound (obstruction B, $k \geq 18$) together cover all k, with a wide overlap at $18 \leq k \leq 10000$. The Lean formalization of the Junction Theorem is in `lean/skeleton/JunctionTheorem.lean` and establishes the logical framework (case split on k, Simons–de Weger axiom, nonsurjectivity chain).

7.2 Open Problems

(a) **Convergence.** Every positive orbit reaches 1?

(b) **Negative cycles.** Nontrivial negative cycles?

(c) **Full Lean formalization.** Baker–Wüstholz as a Lean theorem (not axiom)?

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